Product Introduction

An adaptive learning system for final exam preparation. Built for international students studying in English-medium environments.

6-Layer Modular Architecture

From Unstructured Content to Behavioral Triggers

Core Entities & Data Models

Concept

The atomic unit of knowledge including ID, Chapter, Importance (from syllabus), and Dependency relations.

ConceptState

Dynamic modeling of Mastery [0-1], Confidence, Avg. Response Time, and specific Error Patterns.

StudyPlan

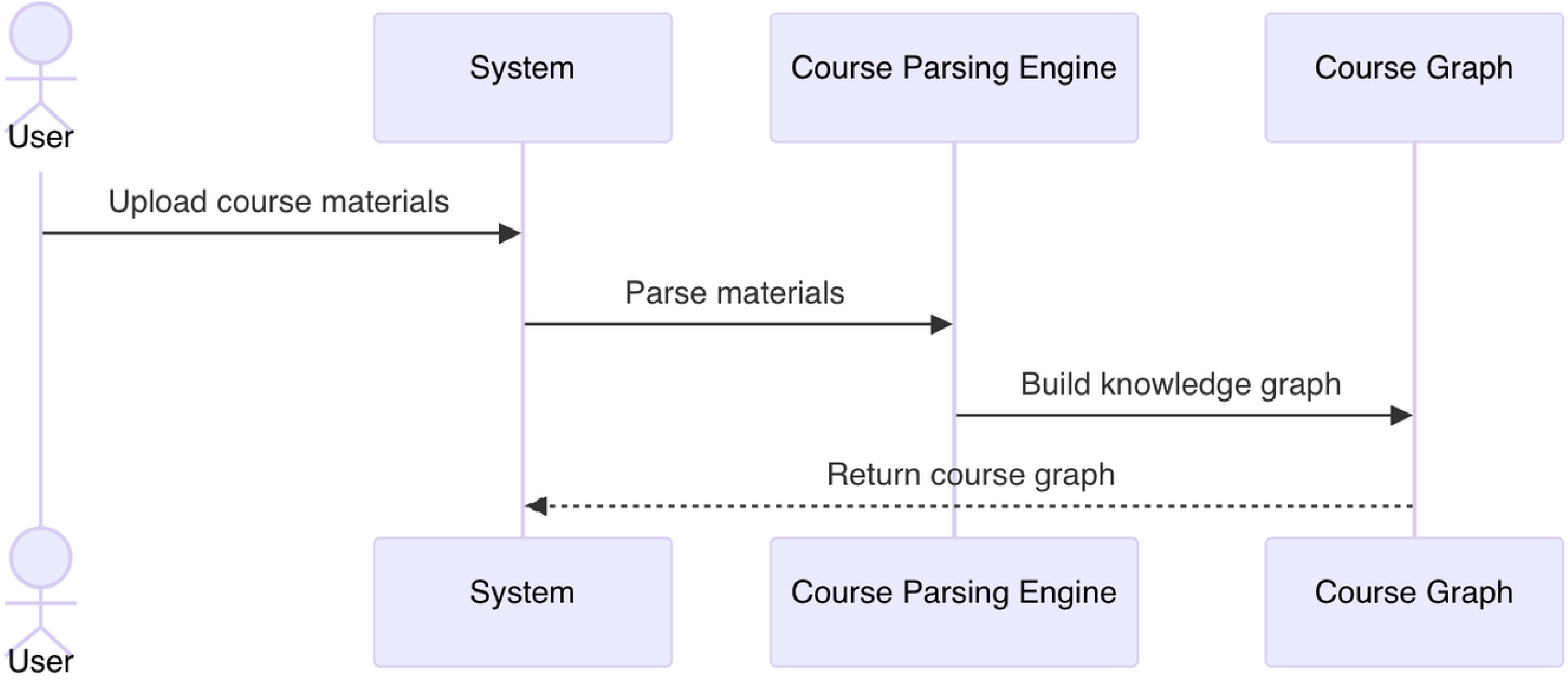
Chronological task queue mapping Concepts to Action Types (Learn/Review/Test) based on constraints.

Course Parsing Engine

Structuring Knowledge

Translates PDFs/PPTs into a Knowledge Graph using rule-based and NLP techniques.

- **Technique:** TF-IDF / TextRank for candidate keyword extraction.
- **Dependency:** Sequence-based rules (Order of slides) + Logical Prerequisites via Semantic Similarity.
- **Metrics:** Precision/Recall of concept extraction; >95% coverage of original materials.



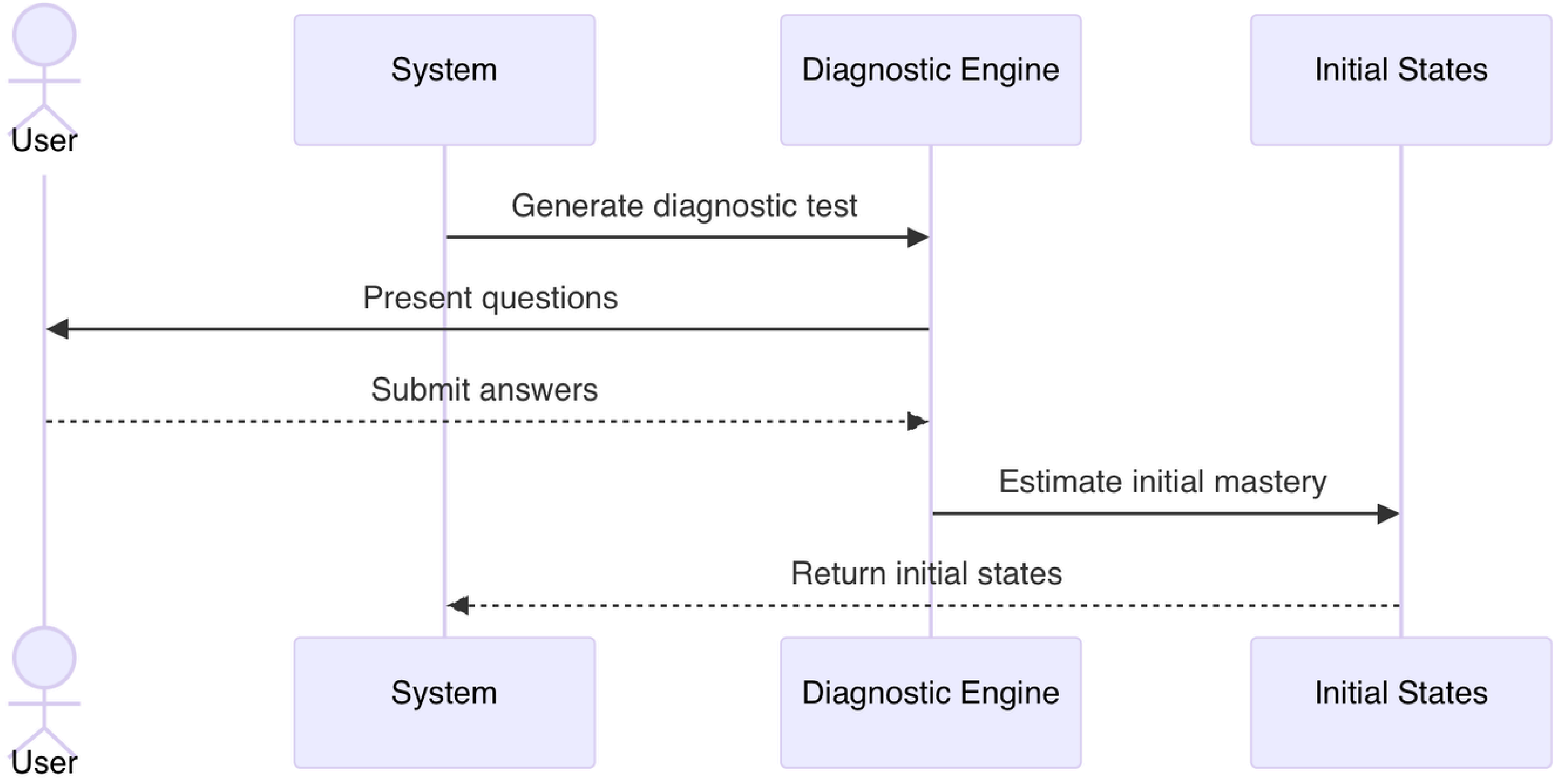
Diagnostic & Assessment Strategy

Strategic Coverage

For every concept, the engine generates three levels of questions: **Definition, Logic/Reasoning, and Application.** This ensures multi-dimensional state estimation.

Non-AI Dispatching

Scheduling follows a breadth-first traversal of the course graph to identify broad gaps before deep-diving into specific chapter failures.



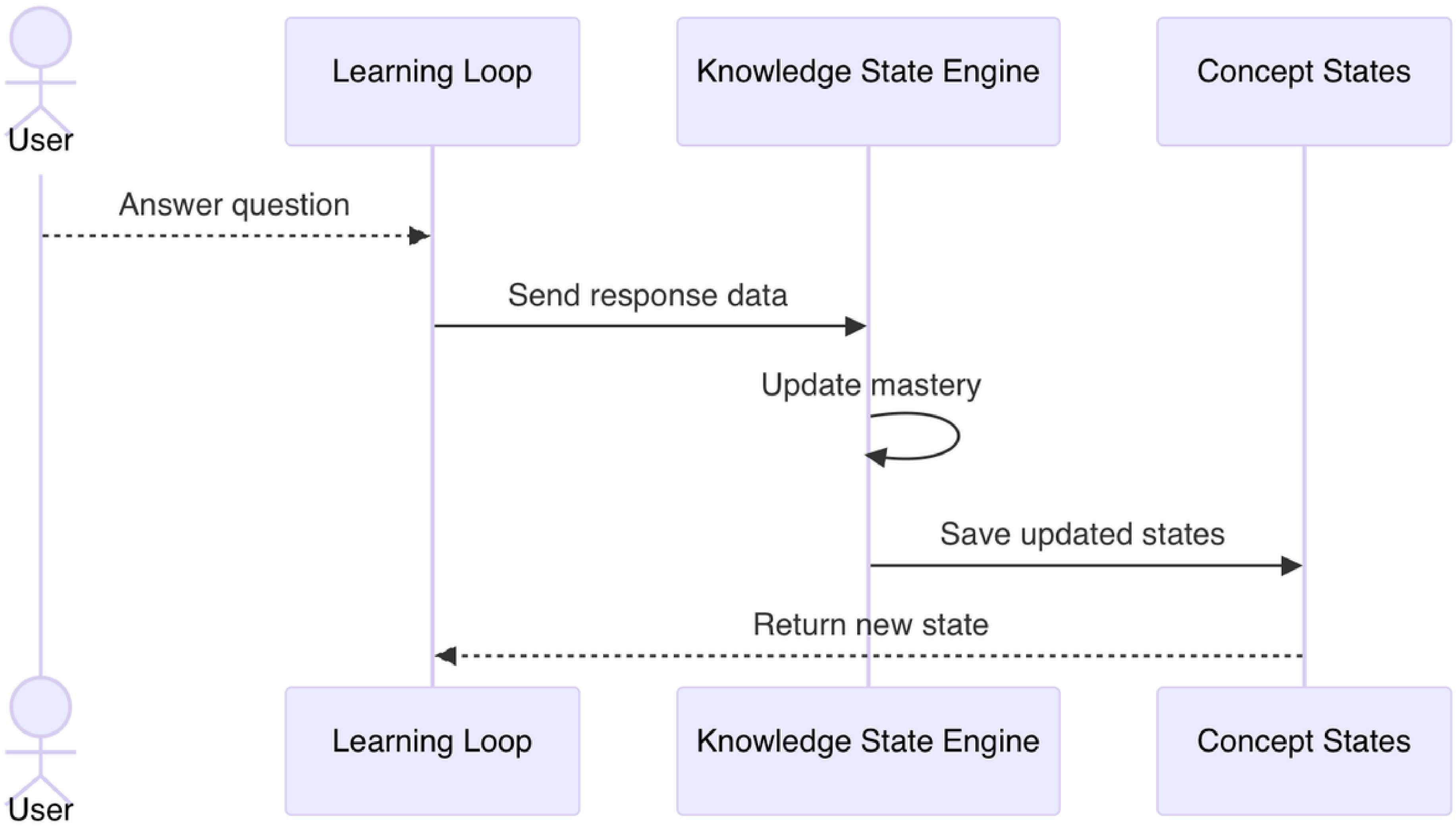
Knowledge State Engine Logic

Mastery Modeling Path

Instead of simple scoring, mastery is a probability calculated via a state-update function. It accounts for noise (lucky guesses) and slips (careless errors).

$$M_{new} = \text{clamp}\,(\,M_{old} + \Delta_{ans} - \Delta_{time} - \Delta_{conf}\ ,\ 0\ ,\ 1\,)$$

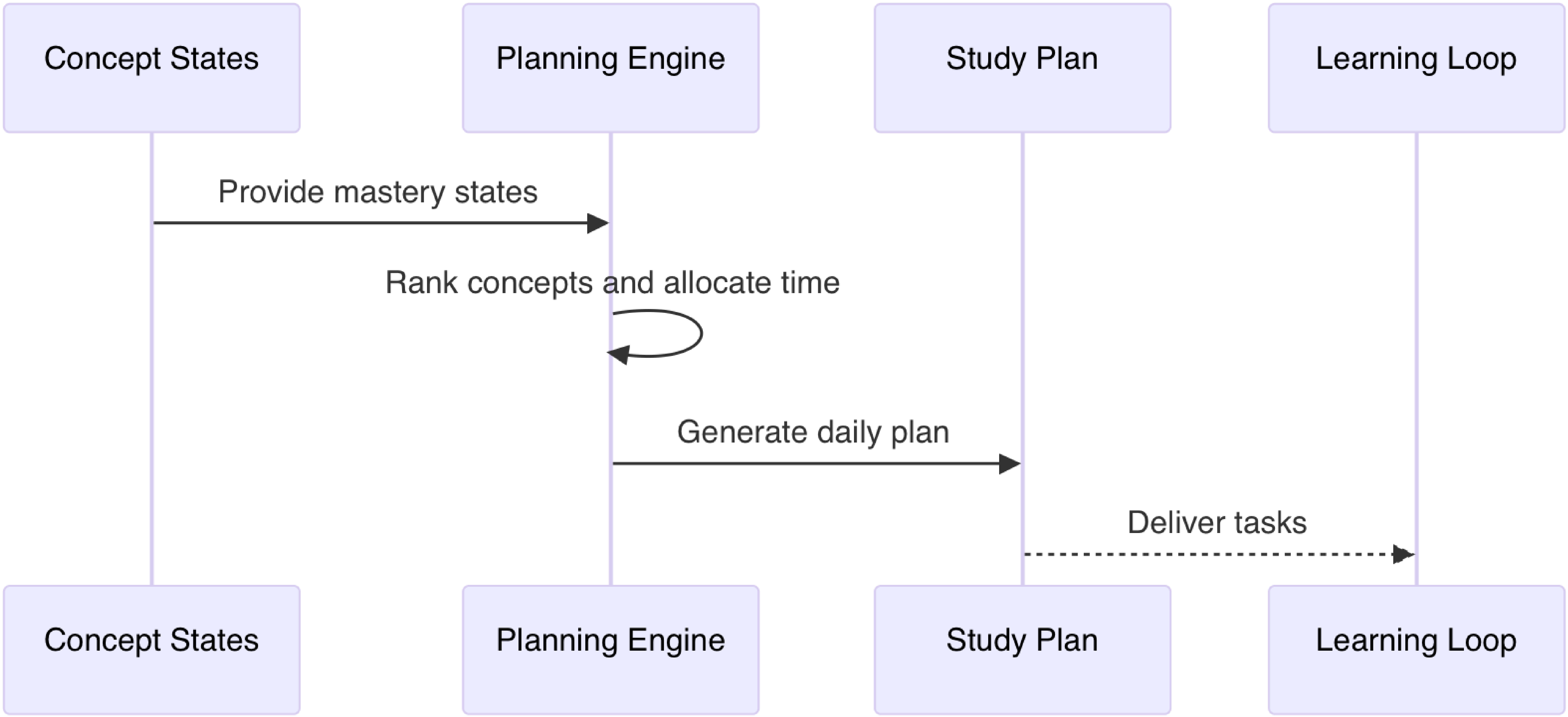
- **Correct Answer:** +0.1 Mastery
- **Fast Wrong:** -0.2 (Knowledge Gap)
- **Slow Correct:** +0.05 (Low Fluency)
- **Low Confidence:** Penalty factor (Guessing detection)
- **AUC Metric:** Predictive accuracy for next-question performance.



Planning Engine: Priority Logic

Scheduling Algorithm: Heuristic search based on Days-Left and Cumulative Difficulty.

$$\text{Priority} = (1 - \text{Mastery}) \times \text{Importance} \times \text{ExamWeight}$$



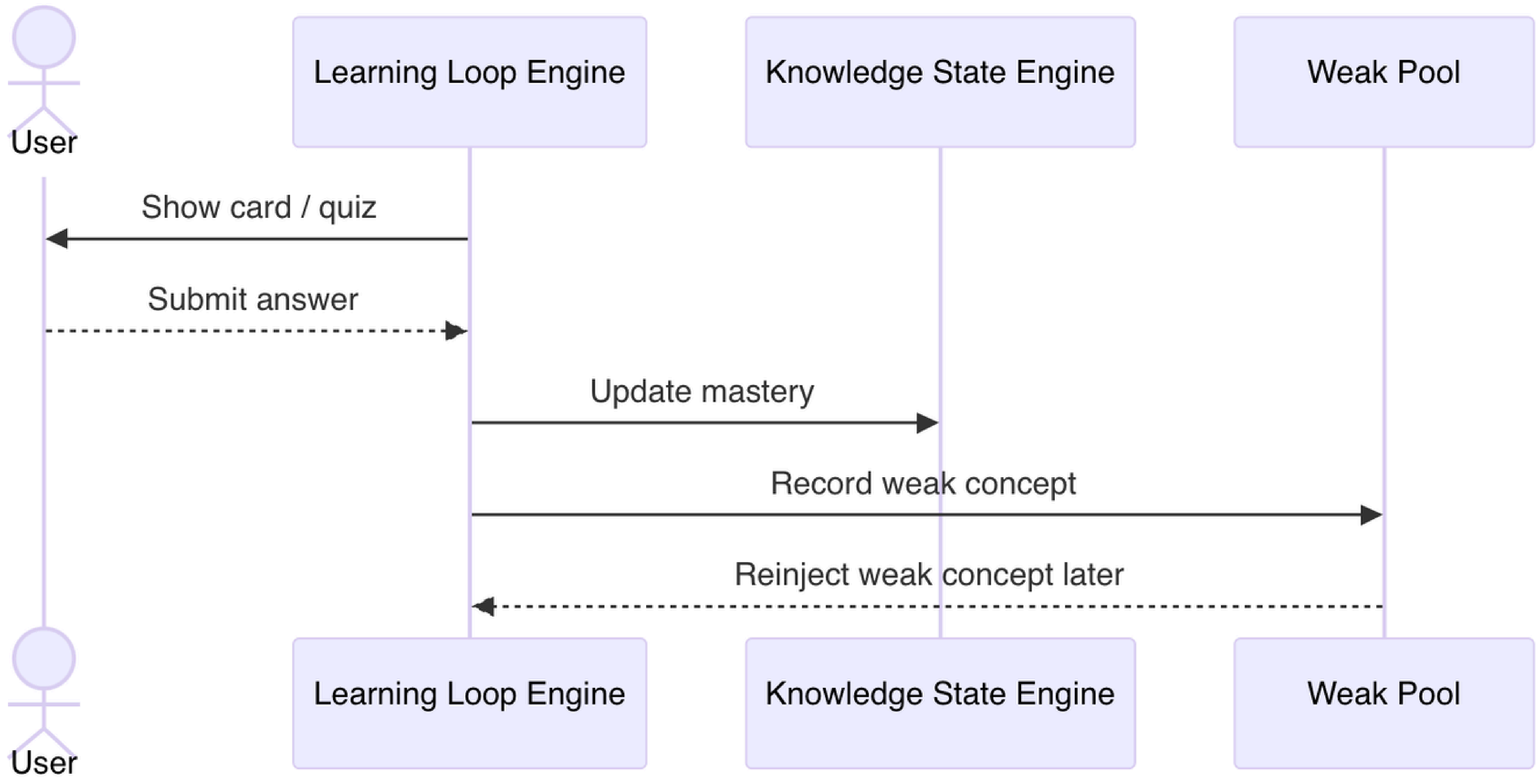
Learning Loop Cycle

The Active Recall Circuit

Learning is structured as a repeating circuit: **Card → Quiz → Feedback → Weakness Pool → Re-injection.**

Every 3 cards, the system forces a retrieval test. If failed, the concept is flagged and re-entered into the planning queue for next-day priority.

Non-AI Mechanism: Spaced Repetition (SRS) scheduling based on Ebbinghaus forgetting curves tailored to student response latency.



Behavior Activation Engine

Temporal Analysis

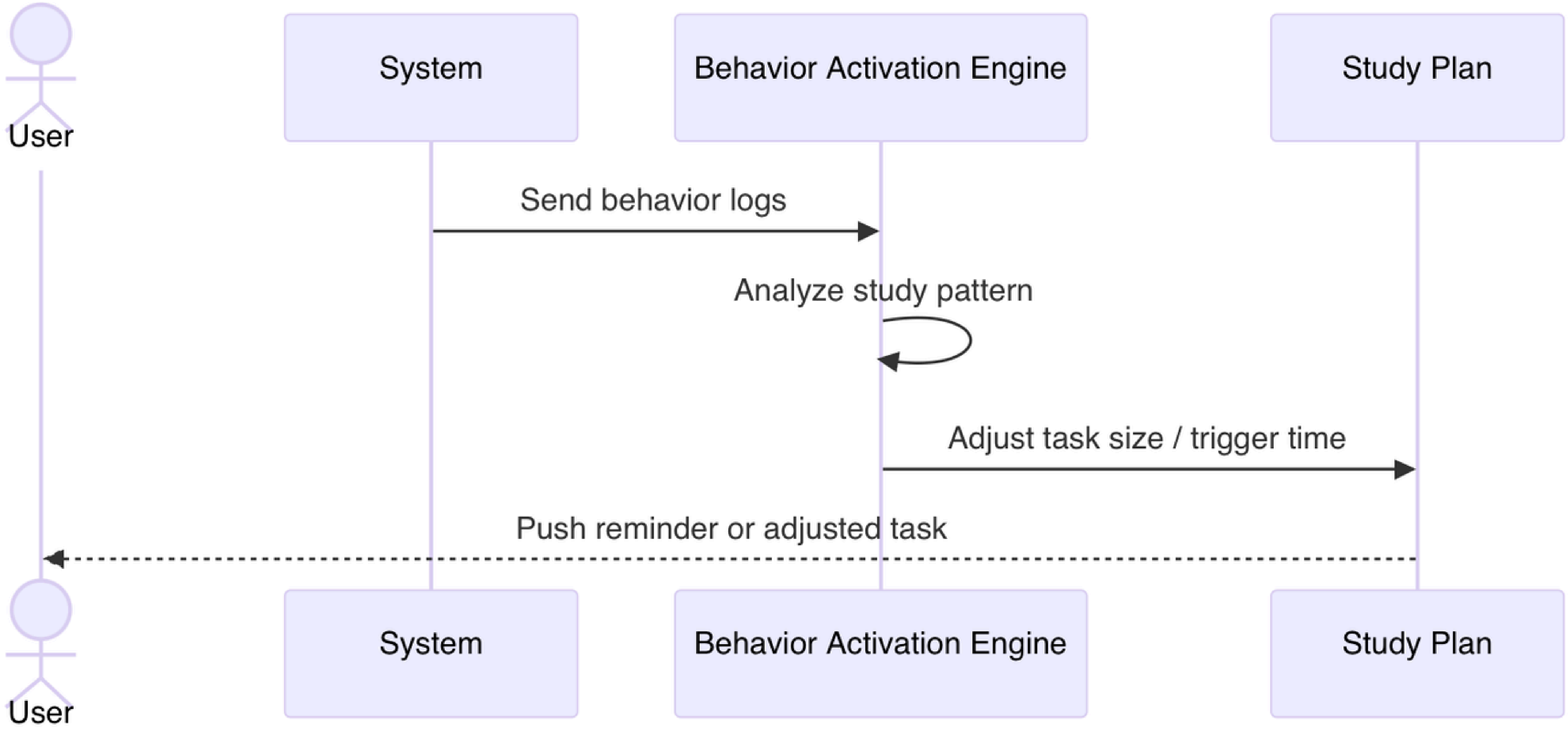
Analyzes peak performance windows. High-difficulty tasks are pushed during peak hours; micro-tasks (5m) during fragmented time.

Trigger Strategy

Custom notifications based on inactivity thresholds. Logic: IF (Mastery_Drop > 10%) THEN (Trigger Remediation).

Adaptive Friction

Adjusts task length and UI complexity based on user fatigue levels detected through response time volatility.



System Technical Specifications

Component	Inputs	Outputs	Measurable Specs
Course Parsing	Syllabus, Slides, PDFs	Course Knowledge Graph	P/R Accuracy; Coverage >95%
Diagnostic Engine	concepts, difficulty config	questions + userResponses	≥2 questions per concept; balanced difficulty; quality rated.
State Modeling	User Response + Latency	Mastery Probability Vector	Prediction AUC; Convergence Rate
Planning Engine	State Vector + Exam Date	Chronological Study Path	Plan Completion Rate; Gain/Hour
Learning Loop	Plan Tasks + Card Assets	Feedback + Weakness Pool	Retention Rate; Error Decay
Behavior Engine	Logs + Habit History	Smart Triggers + UI Adjust	D1/D7 Retention; Activation CTR